

ITERATION PREPARATION

**ITERATION 2 PREPARATION - TITLE
GENERATION ALGORITHM**

Starting Conditions and Improvement Plan

PREP-20260114_112745

EXPERIMENT:	Title Generation Algorithm
CYCLE:	2
DATE:	2026-01-14 11:27:45
STATUS:	Ready to Begin
PRIORITY:	High - 5 Key Improvements

 INFO

ITERATION 2 - Same starting conditions, improved algorithm

ITERATIVE IMPROVEMENT PROCESS

Title Generation Algorithm - Iteration 2 Preparation

Prepared: 2026-01-14
Cycle: 2
Status: Ready to Begin

STARTING CONDITIONS (SAME AS CYCLE 1)

Test Cases (12 Variations)

- 1.The feedback system (/love-you and /hate-this commands) has been implemented
- 2.The PDF generator footer now displays AI assistant information (model name and generation date) instead of arXiv 2026
- 3.The title generation algorithm has been improved to create more informative titles
- 4.The user authentication system now supports OAuth2 and JWT tokens
- 5.The database migration script successfully converts legacy data to the new schema
- 6.The API endpoint /api/v1/users returns user data in JSON format with pagination
- 7.The frontend component renders correctly on mobile devices (iOS and Android)
- 8.The error handling mechanism logs all exceptions to a centralized logging service
- 9.The test suite coverage has increased from 60% to 85% across all modules
- 10.The deployment pipeline automatically builds and deploys to production when tests pass
- 11.The caching layer reduces database query time by 70% for frequently accessed data
- 12.The security module now validates all input data and prevents SQL injection attacks

Current Algorithm State

- **File:** scripts/prove_it_comprehensive.py
- **Function:** generate_informative_title()
- **Current Length Limit:** 75 characters
- **CSS:** Updated for multi-line support

TARGET IMPROVEMENTS (FROM CYCLE 1 OBSERVATIONS)

Priority 1: Preserve Metrics and Percentages

Problem: Variation 9 loses "60% to 85%" metrics

Solution: Detect and prioritize numeric data (percentages, numbers, ranges)

Priority 2: Handle Multiple Features

Problem: Variation 4 only shows OAuth2, loses JWT tokens

Solution: Detect "and" conjunctions and include multiple features

Priority 3: Increase Effective Length

Problem: Variations 5, 8, 10, 11, 12 truncate before key information

Solution: Increase to 90-100 chars OR better prioritization of key information

Priority 4: Improve Outcome Specificity

Problem: Variation 3 uses vague "improved"

Solution: Extract more specific outcome descriptions

Priority 5: Better Truncation Strategy

Problem: Cuts off important endings ("new schema", "logging service", "injection attacks")

Solution: Prioritize endings that contain key information

IMPLEMENTATION PLAN

Step 1: Enhance Pattern Matching

- Detect numeric data (percentages, numbers, ranges)
- Detect "and" conjunctions for multiple features
- Extract more specific outcome verbs

Step 2: Improve Prioritization

- Prioritize: metrics > features > outcomes > conditions
- Preserve endings that contain key information
- Better handling of technical terms

Step 3: Increase Length OR Better Compression

- Option A: Increase to 90-100 chars with better CSS
- Option B: Smarter compression that preserves key info

Step 4: Test and Validate

- Run same 12 test cases
 - Compare results to Cycle 1
 - Document improvements
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SUCCESS CRITERIA

Must Have

- ☒ All 12 variations preserve key information
- ☒ Metrics/percentages preserved (Variation 9)
- ☒ Multiple features included (Variation 4)
- ☒ No awkward truncation (Variations 5, 8, 10, 11, 12)

Should Have

- ☒ Better outcome specificity (Variation 3)
- ☒ Grammatical correctness (Variation 7)
- ☒ Titles fit within display constraints

Nice to Have

- ☒ Semantic understanding of importance
 - ☒ Context-aware abbreviation handling
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NEXT STEPS

1. **Modify Algorithm:** Implement priority improvements
 2. **Test:** Run 12 variations again
 3. **Compare:** Document improvements vs Cycle 1
 4. **Iterate:** Continue to Cycle 3 if needed
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Ready to Begin Iteration 2